

A. Course General Information:

Course Code:	CSE260
Course Title:	Digital Logic Design
Credit Hours (Theory+Lab):	3 + 0
Contact Hours (Theory+Lab):	3 + 3
Category:	Program Core
Type:	Required, Engineering, Lecture + Laboratory
Prerequisites:	CSE250, CSE251
Co-requisites:	None

B. Course Catalog Description (Content):

This course provides an introduction to digital systems such as computers, communication and information systems. Firstly, the course will cover Boolean algebra, digital logic gates, combinational logic circuits, decoders, encoders, multiplexers and demultiplexers. The course will then cover sequential circuits: asynchronous and synchronous counters, registers, flip-flops. An introduction to memory elements and registers will also be provided. Hands-on experience will be provided through lab works and lab projects. The course includes a compulsory 3-hour laboratory work each week.

C. Course Objective:

The objectives of this course are to

- Familiarization with different number systems and conversion
- Introduce Boolean logic operation and teach students how to use Boolean Algebra and K-maps to realize two-level minimal/optimal combinational circuits
- Teach students other Boolean simplification methods such as Tabulation
- Expose students in the introductory design process of combinational circuits including MSI circuits
- Teach basic operation and analysis of sequential circuits using latches, flip-flops, counters, registers and memory elements.

D. Course Outcomes (COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	Use the knowledge of Number systems and Boolean algebra to perform arithmetic and boolean calculations and build circuits	10%
CO2	Use the knowledge of minimization techniques such as Karnaugh map and tabulation method to realize minimal/optimal combinational circuits along with calculating SOP and POS	30%
CO3	Design and analyze different combinational and sequential circuits such as adder, comparator, encoder, decoder, multiplexer, demultiplexer, counter, register, memory, etc.	40%
CO4	Operate laboratory equipment build, and troubleshoot simple combinational and sequential circuits.	15%
CO5	Work in a team and communicate effectively	5%

E. Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Use the knowledge of Number systems and Boolean algebra to perform arithmetic and boolean calculations and build circuits	c	Cognitive	Lectures, notes, lab class	Quiz, Exam
CO2	Use the knowledge of minimization techniques such as Karnaugh map/tabulation method to realize minimal/optimal combinational circuits along with calculating SOP and POS	c	Cognitive	Lectures, notes, lab class	Quiz, Exam
CO3	Design and analyze different combinational and sequential circuits such as adder, comparator, encoder, decoder, multiplexer, demultiplexer, counter, register, memory.	c	Cognitive	Lectures, notes, lab class	Quiz, Exam, Design Project
CO4	Operate laboratory equipment build, and troubleshoot simple combinational and sequential circuits	e	Psychomotor	Lab Class	Lab Work, Design Project
CO5	Work in a team and communicate effectively	l,j	Affective	Lab Class	Lab work, Design Project

F. Course Materials:

i. Text and Reference Books:

Sl .	Title	Author(s)	Publication Year	Edition	Publisher	ISBN
1	Digital Design	M Morris Mano & M D Ciletti	2012	5 th ed.	Pearson Education	ISBN-13: 978-0-13-277420-8
2	Digital Design: Principles and Practices	J F Wakerly	2005	4 th ed.	Prentice Hall	ISBN-13: 978-0131 863897

ii. Other materials (if any)

- Lecture notes and presentation slides
- Lab hand-outs
- Lab usage manual

G. Lesson Plan:

No	Topic	Week/Lecture#	Related CO (if any)
	Introduction, review of number systems, and binary arithmetic	Lecture 1-3	CO1
	Introduction to boolean algebra and simplification	Lecture 4-5	CO1
	SOP, POS and boolean functions minimization techniques (K-map, Tabulation)	Lecture 6-9	CO2
	Adder-Subtractor	Lecture 10-11	CO3
Midterm			
	Combinational circuit design using Decoder, Encoder, Multiplexer	Lecture 13-14	CO3
	Introduction to sequential circuit design process (Flip Flop)	Lecture 15-16	CO3
	Sequential circuit analysis and counter applications	Lecture 17-19	CO3
	Register	Lecture 20-21	CO3
	Memory	Lecture 22	CO3
	Review	Lecture 23	
Final Exam			

(** Please insert separate table of lesson plan for lab component if there is any)

H. Assessment Tools:

Assessment Tools	Weightage (%)
Attendance	5
Assignment	5
Quizzes	15
Midterm Exam	25
Final Exam	30
Project work	5
Lab work (Hardware)	15

I. CO Assessment Plan:

Assessment Tools	Course Outcomes				
	CO1	CO2	CO3	CO4	CO5
Quizzes (Q)	x	x	x		
Assignments	x	x	x		
Midterm exam	x	x	x		
Final Exam			x		
Project work					x
H/W Lab work				x	

J. CO Attainment Policy:

As per Department of CSE Course Outcome Attainment Policy

K. Grading policy:

As per Brac University grading policy

L. Course Coordinator: Dewan Ziaul Karim (DZK)